

2016 Level II Mock Exam: Morning Session

The morning session of the 2016 Level II Chartered Financial Analyst® Mock Examination has 60 questions. To best simulate the exam day experience, candidates are advised to allocate an average of 18 minutes per item set (vignette and 6 multiple choice questions) for a total of 180 minutes (3 hours) for this session of the exam.

Questions	Topic	Minutes
1-6	Ethical and Professional Standards	18
7-12	Ethical and Professional Standards	18
13-18	Economics	18
19-24	Financial Reporting and Analysis	18
25-30	Financial Reporting and Analysis	18
31-36	Corporate Finance	18
37-42	Equity	18
43-48	Fixed Income	18
49-54	Fixed Income	18
55-60	Alternative Investments	18
	Total:	180

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Ethics – Kingfisher

The government of a developing country published a request for proposal (RFP) for the development of policies to improve the business conduct of its capital markets licensees, with the hope of improving confidence levels among investors.

Kingfisher Financial Development Partners responded with a detailed proposal including the following justifications for why the firm should win the tender:

Justification 1: With a team of three CFA charterholders, Kingfisher is more qualified than our competitors to design policies to uphold and enhance capital market integrity.

Justification 2: Each team member must annually renew his or her commitment to abide by the CFA Institute Code of Ethics and Standards of Professional Conduct (Code and Standards).

Justification 3: In addition, every team member passed each level of the CFA exam on the first attempt.

Kingfisher is later notified that it had won the tender. The Kingfisher team consists of team leader Khalid Juma, CFA, and his two associates, Vimal Bachu, CFA, and Anila Patel, CFA. Kingfisher and the government agree that the first step toward improving market integrity is to create an industry-wide code of conduct based on the Code and Standards. Although the Code and Standards are not intended to be adopted in full by the government, the decision is made to concentrate on four main areas: professionalism, capital market integrity, duties to clients, and investment recommendations.

The Kingfisher team subsequently drafts the following policy statements:

Levels of Professionalism

Financial services professionals must act in a professional manner at all times to help protect the integrity of the country's capital markets. As such, financial services professionals must ensure that they meet at a minimum three major requirements. Professionals must (1) disclose all conflicts of interest, (2) selectively differentiate services to clients, and (3) outline all manager compensation arrangements for clients.

Capital Market Integrity

Financial services professionals must protect the integrity of the capital markets by ensuring that any insider information obtained is managed in such a way as to prevent the investing public from being disadvantaged. In addition, no financial services professional can knowingly participate in any activity devised to mislead investors or distort any price-setting mechanism.

Duties to Clients

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Clients' interests must come before those of the financial services firm and/or its staff. To ensure that clients' interests are protected, all portfolios must be invested according to each client's investment plan and must be well diversified across all asset classes available. Furthermore, fund managers must annually review client needs and objectives and rebalance portfolios if required.

Investment Recommendations

All investment recommendations should be made after extensive research undertaken by or on behalf of the firm. In addition, each research report must

Requirement 1: be reviewed by peers as soon as practical to ensure adequate basis and due diligence policies were followed,

Requirement 2: be assessed to determine the quality of the recommendation over time, and

Requirement 3: only include names of team members who took part in the research and agreed with the recommendation.

The Kingfisher team and the government committee meet to agree on the draft code of conduct. Members of the government committee suggest the following additional policy: "Each financial services firm must have a compliance supervisor to ensure that

Task 1: systems are in place to detect violations of laws, rules, regulations, firm policies, and the industry-wide code of conduct and to enforce investment-related compliance policies;

Task 2: the firm has adequate documented compliance policies and procedures and it trains all personnel on the same and makes sure the policies and procedures are followed; and

Task 3: inadequate procedures are identified and recommendations to correct inadequate procedures are submitted to senior management for approval and implementation."

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1. Which of Kingfisher's statements in the RFP regarding its qualifications *most likely* violates the CFA Institute Standards of Professional Conduct?
 - A. Justification 1.
 - B. Justification 2.
 - C. Justification 3.

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2. With regard to the proposed policy statement relating to Levels of Professionalism, which draft requirement *least likely* reflects any of the CFA Institute Standards of Professional Conduct?
- A. Compensation arrangements
 - B. Differentiation of services
 - C. Conflicts of interest
3. Do Kingfisher's proposed policy statements related to Capital Market Integrity *most likely* violate any CFA Institute Standards of Professional Conduct?
- A. Yes, with regard to material nonpublic information
 - B. Yes, with regard to market manipulation
 - C. No
4. Which of Kingfisher's proposed requirements to ensure Duties to Clients is *least* appropriate to prevent violations of CFA Institute Standards of Professional Conduct? The requirement calling for a(n):
- A. diversified portfolio.
 - B. investment plan.
 - C. periodic review.
5. Which of Kingfisher's proposed requirements regarding investment recommendations is *most* appropriate to prevent violations of Standard V(A): Diligence and Reasonable Basis?
- A. Requirement 1
 - B. Requirement 3
 - C. Requirement 2
6. Which of the following tasks suggested by the government committee would *least likely* conform to Standard IV(C): Responsibilities of Supervisors?
- A. Task 1
 - B. Task 2
 - C. Task 3

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Ethics – Adam

Nine months ago, Makenna Adam, CFA, was dismissed from her job as an equity research analyst with Transcontinental Brokerage Company, a publicly listed nationwide stock brokerage company. Unable to find new employment, Adam establishes an Internet-based business, Adam Research Ltd., selling research reports to individuals, institutional investors, and sell-side financial services companies.

Adam recognizes that she must make numerous disclosures on her website to comply with the CFA Code of Ethics and Standards of Professional Conduct (Code and Standards) as well as the CFA Institute Research Objectivity Standards. She believes it is important to comply with the Code and Standards to help improve her business prospects. Adam clearly displays the following claim on the home page of Adam Research's website:

Adam Research Ltd. complies with the CFA Institute Research Objectivity Standards. Investors can be assured that all research is accurate, although actual outcomes may differ from forecasted outcomes. Our research reports clearly distinguish between facts and opinions by the analyst writing the research report. Analysts are also free to voice their own opinions when making recommendations without fear of reprisal to ensure their independence.

Also clearly displayed on the home page is an additional disclosure regarding potential conflicts of interest:

Adam Research Ltd. and/or its employees and associates may occasionally hold shares in any of the companies we cover. Please contact us for disclosure concerning our share positions.

In addition, Adam creates a stock rating system, again posting it on the website for her clients and potential clients so they can understand the basis for how Adam Research recommendations are made. She describes the rating system as follows: The firm uses different recommendation categories (outperform, neutral, and underperform), along with an indication regarding risks for each type of investor and the time frame in which the shares are expected to reach their target price.

Adam realizes she must produce research reports quickly to have product to sell. Her first report covers her former employer, Transcontinental, and is based in part on last year's annual report. Because she is a former employee and a shareholder in Transcontinental, Adam is convinced she knows all aspects of the company very well and decides not to meet with Transcontinental management. She publishes the report, clearly stating she is a former employee and current shareholder. To drive traffic to her website, she allows free access to the report, leaving it on the site even after Transcontinental reports its year-end financial results. She receives an excellent response, with roughly 45% of her marketing list downloading the report.

The Transcontinental report captures investors' attention because of its strong "buy" recommendation, in contrast to other analyst reports recommending a "sell." As a result, Adam is invited to participate in an interactive internet chat room discussion during which she recommends a "buy" for Transcontinental. Because of limited time, she discloses only her former employment at

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Transcontinental and uses the rest of the time to advertise Adam Research. On several occasions, Adam mentions her website's URL.

To expand Adam Research's research capability after obtaining new clients, Adam hires two analysts. Recognizing the need to have written implementation policies because Adam is no longer the only one writing research reports, she creates policies and provides them to the new employees before posting them on the Adam Research website for clients to download. These policies are provided in Exhibit 1.

Exhibit 1

Adam Research Company Policies and Procedures

Policy Type	Document Content Description
Research Objectivity Policy	This document describes the process required to determine whether there is independence and objectivity in the firm's research, with instructions to make this policy available to all investors and employees. Procedures cited include supervisory procedures to ensure compliance, annual attestation, and adherence to internal audit requirements.
Compliance and Enforcement Policies	This document describes compliance policies and procedures to ensure research objectivity and lists all activities considered to be violations and the resulting disciplinary sanctions, including dismissal from the firm.
Personal Investments and Trading Policies	This document describes the policies designed to manage covered employees' personal investments and trading activities to ensure the interests of the clients are always placed before the company, its employees, and their immediate families, including prohibition of front running and participation in subject company IPOs. In addition, covered persons are banned from trading against the company's recommendations unless for financial hardship reasons. All trades must be approved in advance.

7. Which of the following initial claims made on the home page of Adam Research's website *least likely* reflects the CFA Institute Research Objectivity Standards?
- A. Distinction between fact and opinion
 - B. Independence of analysts' recommendations
 - C. Accuracy of research reports

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8. Adam Research's website disclosure regarding potential conflicts of interest *least likely* meets the recommendations for compliance with the CFA Institute Research Objectivity Standards concerning the:
- A. plain language.
 - B. comprehensiveness.
 - C. prominent display.
9. Which category of Adam Research's stock rating system could *most likely* be improved to meet the recommendations for compliance of the CFA Institute Research Objectivity Standards?
- A. Recommendation categories
 - B. Investor risk
 - C. Time horizon
10. The research report on Transcontinental *most likely* meets recommendations for compliance with the CFA Institute Research Objectivity Standards with regard to:
- A. reasonable and adequate basis.
 - B. timeliness of research reports and recommendations.
 - C. relationships with subject companies.
11. Did Adam's participation in an interactive internet chat room discussion *most likely* comply with recommendations for compliance of the CFA Institute Research Objectivity Standards and Standards of Professional Conduct?
- A. No, because she is trying to manipulate the share price
 - B. Yes
 - C. No, because she did not make sufficient disclosures
12. Which of Adam Research's mentioned company policies and procedures given in Exhibit 1 *least likely* complies with the CFA Institute Research Objectivity Standards?
- A. Compliance and Enforcement
 - B. Personal Investments and Trading
 - C. Research Objectivity

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Economics – Drawbridge

Charles Hollingsworth is an investment strategist at Drawbridge Asset Partners (Drawbridge), an international investment firm. He is meeting with equity analyst Andrew Gillibrand and fixed-income analyst Eliana Navarro to discuss new investment opportunities and the economic factors they should consider as they make their investment selections.

Hollingsworth begins the meeting with the following statement:

"Before we look at new investment opportunities, I want to review some prior transactions. A few months ago, Drawbridge entered into a carry trade in a set of currencies. This morning, we were unfortunately forced to close out the position at a sizable loss as a result of unexpected market volatility."

Hollingsworth continues:

"Earlier in the year, Drawbridge hedged a long exposure to the Australian dollar (AUD) by selling AUD5 million forward against the US dollar (USD); the all-in forward price was 0.8940 (USD/AUD). It is now three months prior to the settlement date, and I want to mark the forward position to market."

Exhibit 1 provides information about current rates in the foreign exchange markets.

Exhibit 1	
Current Foreign Exchange Data	
Spot rate (USD/AUD)	0.9062/0.9066
Three-month points	-36.8/-36.4
Three-month Libor (AUD)	2.88%
Three-month Libor (USD)	0.23%

On completion of the agenda items relating to the foreign exchange markets, Hollingsworth and his team move on to new investment opportunities. They begin with a discussion about the relationship between economic growth and the performance of equity and debt markets.

Gillibrand: "When we consider our equity investments over the long term, our primary focus should be on the rate of GDP growth. For longer time horizons, changes in earnings and the price/earnings multiple are relatively less important in determining appreciation in the stock market."

Navarro: "When we look at our fixed-income investments, we should keep in mind that higher rates of potential GDP growth will translate into higher real interest rates and higher expected real asset returns."

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Hollingsworth: "Anticipating changes in potential GDP can be quite lucrative for us because credit rating agencies often use the growth of potential GDP as an input in evaluating sovereign risk. In general, there is an inverse relationship between estimated potential GDP growth and credit quality."

The economic growth projections for two of the countries in which Drawbridge is considering making new investments are presented in Exhibit 2. Hollingsworth prefers the Solow growth accounting equation to calculate potential GDP growth rather than the more simplistic labor productivity growth accounting equation.

Exhibit 2					
Long-Term Growth Projections					
Country	Inflation Rate (%)	Growth in Total Factor Productivity (%)	Output Elasticity of Capital	Growth Rate of Capital (%)	Growth Rate of Labor (%)
Country A	1.7	1.5	0.3	3.2	0.4
Country B	1.8	1.3	0.4	3.7	0.5

The conversation then turns to the topic of convergence. Navarro says: "Even though Country B's per capita growth is expected to exceed that of Country A for some time, according to the neoclassical model, eventually both countries will experience the same growth rate because the model assumes all countries have access to the same technology."

Hollingsworth presents the long-term relative performance of Countries C and D, shown in Exhibit 3. Although both countries had below-average levels of per capita GDP 50 years ago, over time, the per capita GDP growth rate of Country C has risen rapidly and for nearly 20 years has been well above average. The growth rate for Country D, however, has risen more slowly. Today, Country C ranks among the advanced economies whereas Country D remains a developing nation.

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Exhibit 3			
Real Per Capita GDP Growth			
Country	GDP/Capita 50 Years Ago	GDP/Capita Today	GDP Growth Rate Over Past 50 Years
Country C	6,950	35,190	3.30%
Country D	8,240	20,410	1.83%

13. The primary factor that was *most likely* the cause of Drawbridge's outcome in its carry trade was:
- A. flight to safety.
 - B. leverage.
 - C. stop-loss orders.
14. The mark-to-market value for Drawbridge's forward position is *closest* to:
- A. –USD42,576.
 - B. –USD44,774.
 - C. –USD44,800.
15. Which of the statements about economic growth and the performance of equity and debt markets is the *least* accurate?
- A. Hollingsworth's
 - B. Navarro's
 - C. Gillibrand's
16. Based on the data in Exhibit 2, the GDP growth rate in Country A using Hollingsworth's preferred method of calculation is *closest* to:
- A. 2.94%.
 - B. 2.74%.
 - C. 2.86%.
17. Navarro's statement about the convergence of growth between Country A and Country B is *best* described as:
- A. conditional convergence.
 - B. absolute convergence.

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C. club convergence.

18. Country D's current economic status can *best* be explained by past government policies that encouraged:

- A. foreign investment.
- B. domestic substitutes.
- C. free trade.

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Mark Crawley is an analyst at a London-based private equity firm and is reviewing the firm's file on Thames Air Plc (Thames), a company it provides financing for. Thames uses International Financial Reporting Standards (IFRS) in the preparation of its financial statements. Thames is a relatively new airline based in the United Kingdom specializing in flights and vacation packages to Mediterranean locations, primarily Spain. Thames sells most of its flights and vacation packages to British residents in British pounds (GBP) and considers the costs of local competitors' packages when determining its prices. Costs are incurred in multiple currencies:

- Wage costs are primarily in GBP.
- Typical of the industry, airline fuel and lease costs are normally priced in US dollars (USD).
- The landing fees paid at the vacation-area airports are in the local currency, primarily euros (EUR).

First, Crawley turns his attention to the effect of the transactions undertaken in various currencies by Thames.

- He reviews the change in the exchange rate for the USD to GBP during 2015, shown in Exhibit 1, and wonders what the effect of this change was on Thames's operating income.
- At year-end (31 December), Thames had a large outstanding payable in Spain related to landing fees that were incurred there evenly over the final quarter. The company paid the amount in full on its due date of 28 February. Crawley observed that the EUR to GBP exchange rate had changed between when the costs were incurred and the year-end and again by the payment date, as also shown in Exhibit 1.

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Exhibit 1		
Selected Exchange Rate Data		
	GBP/USD Close	GBP/EUR Close
1-Jan-15	0.64	
30-Jun-15		0.72
31-Dec-15	0.68	0.75
28-Feb-16		0.73
Average, 1 July-31 December 2015		0.7325
Average, 1 October-31 December 2015		0.74

Because of the growing demand for vacation rentals in Spain during the past year, Thames acquired 100% of Tagus SA (Tagus), a Spanish company that owns a small vacation hotel and a few villas. Tagus has long-term debt outstanding from a Spanish bank that financed the 2012 purchase of the vacation properties, which will now be rented as part of the vacation packages offered by Thames. Tagus incurs all costs related to operating and maintaining the rental properties in EUR.

Since the acquisition, all of Tagus's revenue comes from Thames's sales in Britain of the vacation packages. Tagus receives the amounts in GBP. But Tagus hopes to expand and start renting out any excess capacity of the properties, or newly acquired properties, to local tourists in the next few years. Crawley notices that Thames is using the temporal method to translate Tagus's financial statements prior to consolidation and asks another analyst, Dee Chopra, if this is appropriate.

Crawley next reviews the information in Exhibit 2 related to the Tagus acquisition to consider the effect on Thames's year-end financial statements (31 December 2015).

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Exhibit 2
Selected Financial Information of Tagus SA
at Acquisition and Year-End

(EUR thousands)	Balance Sheet Date of Acquisition (30 June 2015)	Balance Sheet Year-End (31 December 2015)	Income Statement for Six-Month Period Ending 31-Dec-15	
Cash and accounts receivable	4,000	4,200	Revenues	8,200
Inventory	2,000	2,250	Operating costs	<u>6,485</u>
Capital assets	<u>15,000</u>	<u>14,625</u>	Operating income	1,715
Total assets	21,000	21,075	Interest expense	<u>395</u>
			Earnings before taxes	1,320
Current liabilities	3,500	3,400	Income taxes	<u>395</u>
Long-term debt	10,000	9,750	Earnings after tax	<u>925</u>
Share capital	5,000	5,000	Dividends	500
Retained earnings	<u>2,500</u>	<u>2,925</u>		
Total liabilities and shareholders' equity	21,000	21,075		

As the final step in his review, Crawley starts a ratio analysis of Thames and Tagus, and he asks Chopra which ratios, if any, would be unaffected by Thames's choice of translation method for Tagus.

19. The functional currency of Thames is *most likely*:

A. USD.

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- B. GBP.
 - C. EUR.
20. Which of the following statements about the effect of the change in the USD to GBP exchange rate during the year is *most* accurate? Operating income for Thames would:
- A. increase because of the positive effect on revenues.
 - B. increase because of the positive effect on operating costs.
 - C. decrease because of the negative effect on operating costs.
21. Which of the following *best* describes the effect on Thames's financial statements of the payment terms related to the landing fees in Spain? Thames would:
- A. defer recognizing any currency effects until the payable is paid.
 - B. report an unrealized exchange loss at year-end.
 - C. adjust the landing fees expense to reflect the change in exchange rate when they are paid.
22. Chopra's *best* answer to Crawley's question about Thames's use of the temporal method to translate Tagus's financial statements is that it is:
- A. incorrect, because the functional currency of Tagus is EUR.
 - B. correct, because the functional currency of Tagus is GBP.
 - C. correct, if the presentation currency of Tagus's financial statements is GBP.
23. The *most likely* effect of the change in the exchange rate between the EUR and GBP arising from Thames's investment in Tagus in 2015 will be a translation:
- A. loss reported in net income.
 - B. gain reported in net income.
 - C. adjustment reported in other comprehensive income.
24. The *best* answer Chopra can give to Crawley's question about which ratio would be unaffected is the:
- A. current ratio.
 - B. operating profit margin.

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C. receivables turnover ratio.

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FRA - Ready Power

Ready Power Inc. is a manufacturer of high-quality industrial electric generators. Although many companies have been negatively affected by the continued global economic weakness, Ready Power has experienced strong demand for its products, largely as a result of several recent natural disasters and many occurrences of rolling brownouts and blackouts arising from excessive strains on power grids. Although this strong demand has resulted in higher inventory costs in recent years, the company has been able to pass the cost on to customers through higher prices. The company's generators have expected useful lives of about 25 years. The company also normally depreciates its assets on a straight-line basis.

Margo Lenz, CFA, an equity analyst at Livermore Investment Council, is reviewing Ready Power's recent financial statements, which are prepared according to US GAAP.

Exhibits 1 and 2 contain selected portions of the company's statement of operations and statement of financial position, and Exhibit 3 contains selected notes from the company's 2013 financial statements.

Exhibit 1: Ready Power Consolidated Results of Operations

(\$ millions)		
For the Year Ending 31 December	2013	2012
Sales	24,910	21,803
Cost of goods sold	<u>17,729</u>	<u>15,935</u>
Gross profit	7,181	5,868
Net profit	2,122	1,712

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Exhibit 2: Ready Power Consolidated Financial Position

(\$ millions)		
As of 31 December	2013	2012
Cash	318	665
Receivables	8,983	8,381
Inventories	3,811	3,134
Other current assets	<u>744</u>	<u>1,441</u>
Current assets	13,856	13,621
Net property, plant, and equipment	5,311	4,794
Other assets	<u>11,360</u>	<u>9,826</u>
Total assets	<u>30,527</u>	<u>28,241</u>

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Exhibit 3: Ready Power Selected Notes to Consolidated Financial Statements

Note 1. Operations and Summary of Significant Accounting Policies

D. Inventories

Inventories are stated at the lower of cost or market, with cost determined using the last in, first out (LIFO) method.

(\$ millions)	2013	2012
LIFO reserve	1,442	1,407

No LIFO liquidation occurred during 2012 and 2013.

F. Depreciation and amortization

Depreciation of plant and equipment is computed using the straight-line depreciation method.

(\$ millions)	2013	2012
Consolidated depreciation expense	332	235

J. Income taxes: The company's effective tax rate has always been 29%.

Note 10. Property, plant, and equipment (PP&E)

	31-Dec	
(\$ millions)	2013	2012
Land	110	92
Plant and equipment	<u>10,257</u>	<u>9,426</u>

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Total plant and equipment	10,367	9,518
Less accumulated depreciation	<u>-5,056</u>	<u>-4,724</u>
Net PP&E	5,311	4,794

Harold Mays, one of Lenz's assistants, made the following comments about Ready Power's inventory policy:

1. One of the advantages of using LIFO is that it simplifies the accounting process for inventory because it gives the same results for inventory and cost of goods sold whether the company uses a periodic or perpetual inventory system.
2. Another advantage of using LIFO is that it appears to improve the company's cash conversion cycle.
3. One disadvantage with LIFO, however, is that it is more likely that the company will incur inventory write-downs than under the first in, first out (FIFO) method.

Lenz mentioned to Mays that earlier that day, she had seen Bill Jacobs, the CEO of Ready Power, in an exclusive interview on a cable news network specializing in financial news and information. Lenz was particularly interested in the portion of the interview dealing with the company's new program to lease electrical generators. An excerpt from a transcript of the interview is shown in Exhibit 4.

Exhibit 4: Excerpt from an Interview of Bill Jacobs on Cable TV, 4 March 2014

Jacobs: The firm is meeting the growing demand for our electrical generators and will be introducing a leasing program to further consolidate our lead in this area. We anticipate that about 80% of the leases we grant will have a term of 20 years or more, with the remainder having shorter terms of around 5 years.

After reading the excerpt from the interview, Mays wondered what impact the company's new position as a lessor and its classification of leases would have on the company's future financial statements. Finally, he comments:

1. For a given leased asset, in the initial year of the lease, Ready Power's profits should be higher if the company classifies the lease as an operating lease.

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2. Regardless of how the company classifies a lease, its total cash flow and operating cash flow over the lease term will be the same.

3. The leasing program will decrease Ready Power's liquidity position.

25. If Ready Power had used the FIFO method to account for its inventory, its cost of goods sold (COGS) in 2013 would have been *closest* to (in millions):
- A. \$17,694.
 - B. \$17,764.
 - C. \$16,287.
26. If Ready Power had been using FIFO accounting since incorporation, its retained earnings at the end of 2013 would *most likely* be higher by (in millions):
- A. \$2,927.
 - B. \$1,442.
 - C. \$1,024.
27. The statement in Note 1.D of Exhibit 3 concerning LIFO liquidations *most likely* means that for the stated period:
- A. there were no inventory write-downs in either of the two years.
 - B. costs and prices must have been rising throughout.
 - C. units manufactured (or purchased) equaled or exceeded unit sales for each year.
28. With regard to Mays' comments about the LIFO method, which of his statements is *most* accurate?
- A. Statement 2
 - B. Statement 1
 - C. Statement 3
29. In 2013, the estimated remaining life (in years) of the company's asset base is *closest* to:
- A. 16.0.
 - B. 15.2.
 - C. 15.7.
30. Which of May's statements about the new leasing program is *most* accurate?
- A. Statement 1

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- B. Statement 2
- C. Statement 3

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Corporate Finance - Earl Case

John Earl is a project analyst for Kames Inc. Earl is currently reviewing the projected annual income statements, shown in Exhibit 1, for the five-year life of Project #162 to determine the net present value (NPV) of the project using an annual discount rate of 10%.

Exhibit 1
Project #162 Forecasted Income Statements

	Year 1	Year 2	Year 3	Year 4	Year 5
Sales	\$300,000	\$320,000	\$350,000	\$390,000	\$440,000
Cash operating expenses	210,000	224,000	245,000	273,000	308,000
Depreciation	30,000	30,000	30,000	30,000	30,000
Operating income	\$60,000	\$66,000	\$75,000	\$87,000	\$102,000
Interest expense	13,500	10,800	8,100	5,400	2,700
Taxable income	\$46,500	\$55,200	\$66,900	\$81,600	\$99,300
Tax expense (40%)	18,600	22,080	26,760	32,640	39,720
Net income	\$27,900	\$33,120	\$40,140	\$48,960	\$59,580

The project will require an increase in fixed assets of \$150,000 that will be fully depreciated. Current assets are expected to increase by \$80,000 and current liabilities are expected to increase by \$45,000. This increase in net working capital will be recovered when the project is finished.

Just prior to completing the analysis, Earl finds out that the fixed assets can be depreciated using an accelerated method, as shown in Exhibit 2.

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Exhibit 2
Project #162 Forecasted Income Statements with Accelerated Depreciation

	Year 1	Year 2	Year 3	Year 4	Year 5
Sales	\$300,000	\$320,000	\$350,000	\$390,000	\$440,000
Cash operating expenses	210,000	224,000	245,000	273,000	308,000
Depreciation	<u>49,995</u>	<u>66,675</u>	<u>22,215</u>	<u>11,115</u>	<u>0</u>
Operating income	\$40,005	\$29,325	\$82,785	\$105,885	\$132,000
Interest expense	<u>13,500</u>	<u>10,800</u>	<u>8,100</u>	<u>5,400</u>	<u>2,700</u>
Taxable income	\$26,505	\$18,525	\$74,685	\$100,485	\$129,300
Tax expense (40%)	<u>10,602</u>	<u>7,410</u>	<u>29,874</u>	<u>40,194</u>	<u>51,720</u>
Net income	\$15,903	\$11,115	\$44,811	\$60,291	\$77,580

Given the use of the accelerated depreciation method, Earl concludes that the NPV of Project #162 increases to \$127,818 .

In an initial discussion with a fellow analyst, David North, about Project #162, Earl tells North:

“I have prepared the analysis using nominal values and a nominal discount rate.”

North responds:

“Even though the analysis is in nominal terms, the discount rate should be increased by an inflation rate of 2% based on the historical inflation rate.”

Later, Earl and North continue their discussion. Earl explains:

“I intend to also calculate the economic profit by subtracting the dollar cost of capital from the net income. Do you have any further suggestions for analysis?”

North replies:

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"I suggest you determine the key inputs for the analysis, and then examine each input separately by varying its value between plus or minus 1% or 2%. This variance will give you better insight about the project's profitability."

31. Given the information in Exhibit 1, the after-tax operating cash flow (in thousands) for Year 1 for Project #162 is *closest* to:
- A. \$36.0.
 - B. \$66.0.
 - C. \$71.4.
32. The initial investment outlay (in thousands) for Project #162 is *closest* to:
- A. \$275.
 - B. \$185.
 - C. \$230.
33. By switching to an accelerated depreciation method, the increase in NPV for Project #162 is *closest* to:
- A. \$4,445.
 - B. \$11,112.
 - C. \$6,667.
34. In their initial discussion, North's response to Earl is *most likely*:
- A. incorrect because the discount rate does not need to be adjusted.
 - B. incorrect because the inflation rate adjustment should be based on expected inflation.
 - C. correct.
35. Earl's statement in regard to economic profit is *most likely*:
- A. incorrect because it is a residual income calculation.
 - B. incorrect because the calculation should not be based on net income.
 - C. correct.
36. The final suggestion by North is *best* described as:
- A. scenario analysis.
 - B. sensitivity analysis.
 - C. Monte Carlo analysis.

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Equity - Darwin

Gabrielle Marchand and Cristiano Palmeiro are junior analysts recently hired by Nordfjord Investment Management, an international investment firm. They have been assigned by senior analyst Anniken Kristensen to work as a team to research Darwin Industrial (Darwin), a major company in the paints and coatings industry.

Marchand and Palmeiro start by researching the industry. They discuss how the competitive environment could impact profitability and make the following notes:

- The industry is fragmented and there is a strong rivalry for market share, particularly among the larger participants.
- Paints and coatings are the logical or only choice for many applications, but alternatives, such as aluminum, vinyl, and wood, are available for some situations.
- There is some brand loyalty, although it is not pervasive. The essentially identical product offerings from the various manufacturers enable customers to easily switch brands.

Marchand and Palmeiro's research reveals that the industry's growth prospects are predictable because they are closely tied to economic growth, particularly the housing, construction, automotive, and industrial products sectors. Responding to regulatory pressure and increasing consumer demand for environmentally friendly products, Darwin has been at the forefront of developing products that are more eco-friendly and safer for its employees to manufacture and for customers to use.

In developing their sales and expense forecasts, Marchand and Palmeiro review selected financial data on Darwin and selected economic factors, as shown in Exhibit 1. Using 2015 as the base year, the analysts expect Darwin's

- sales to grow 1% faster than projected nominal global GDP growth,
- cost of goods sold to decline 0.5% annually,
- selling expenses to remain stable as a percentage of sales,
- general and administrative and depreciation and amortization expenses to be fixed, and
- net debt to decline €100 million in 2016.

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Exhibit 1 Darwin Industrial Selected Financial Data		
<i>Income statement</i>	2014 (€ millions)	2015 (€ millions)
Sales	8,838	9,280
Cost of goods sold (COGS)	<u>5,183</u>	<u>5,401</u>
Gross profit	3,655	3,879
Selling expenses	1,836	1,940
General and administrative expenses (G&A)	485	485
Depreciation and amortization expenses (D&A)	<u>294</u>	<u>294</u>
Operating profit	1,040	1,160
Interest expense	<u>96</u>	<u>92</u>
Earnings before taxes (EBT)	944	1,068
Income taxes (30%)	<u>283</u>	<u>320</u>
Net profit	661	748
<i>Average balance sheet items</i>		
Total assets		7,730
Net debt		1,533
Total liabilities		4,279
Total equity		3,451
Selected Economic Data		
2016 global GDP growth rate	4.50%	

After completing their forecast of the income statement, Marchand and Palmeiro discuss approaches to forecasting balance sheet accounts. Marchand asks Palmeiro which accounts on the balance sheet can be most reliably forecasted from the income statement.

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Kristensen and her team then move on to a discussion of the various ways of comparing Darwin's profitability with other firms in the industry, and they make the following comments:

Kristensen: I prefer return on invested capital (ROIC) because it is not affected by the amount of debt on Darwin's balance sheet.

Palmeiro: Return on equity (ROE) is the most common measure of shareholder return, although Darwin's share repurchase program will affect the relevance of the ratio.

Marchand: We could use return on capital employed (ROCE), but its significance will be limited if we compare Darwin with companies based in other countries.

-
37. Based on Marchand and Palmeiro's notes, the industry's competitive strength is *most likely* related to the:
- A. rivalry among the firms.
 - B. threat of substitutes.
 - C. bargaining power of buyers.
38. Based on the analysts' research about industry growth prospects and Darwin's response, which of the following would be the *most* appropriate classification of Darwin's strategic style?
- A. Classical
 - B. Visionary
 - C. Shaping
39. Marchand and Palmeiro's modeling approach can be *most* appropriately described as:
- A. top down.
 - B. bottom up.
 - C. hybrid.
40. Based on the analysts' sales and expense forecasts and the data in Exhibit 1, their forecasted net profit for Darwin in 2016 will be *closest* to:
- A. €827 million.
 - B. €853 million.
 - C. €861 million.
41. The *best* answer to Marchand's question about forecasting balance sheet accounts is:
- A. inventory.
 - B. operating loans.
 - C. property, plant, and equipment.

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42. Which of the three analysts' comments about the methods used to compare Darwin's profitability with other firms in the industry is the *least* accurate?
- A. Palmeiro's
 - B. Kristensen's
 - C. Marchand's

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Fixed Income - Wingersheek

Sandy Annisquam is the head of trading at Wingersheek Arbitrage Opportunities, LLP, a hedge fund specializing in fixed income strategies. The firm's investment approach is to exploit small price differences across similar or identical securities. Annisquam has asked Choate Hake to develop a comprehensive automated trading system that will allow traders to identify opportunities in the market. Annisquam and Hake are discussing several applications that need to be developed for the traders.

Hake begins development on an algorithm that will evaluate government bonds that have been stripped. He tests his logic by evaluating a dollar-denominated Tangoran government bond with a 3.20%, annual pay coupon maturing in three years, using data in Exhibit 1. The bond is quoted in the market at \$103.50.

Exhibit 1
Spot, Par and Forward Rates

	Year 1	Year 2	Year 3
Spot Rate	1.10%	1.50%	2.01%
Par Rate	1.10%	1.50%	2.00%
Forward Rate	1.10%	1.91%	3.04%

Hake develops a framework for valuing bonds using a binomial interest rate tree. He understands that there are several factors used in developing the tree and asks Annisquam for counsel on the correct data to use. Annisquam makes the following comments to Hake:

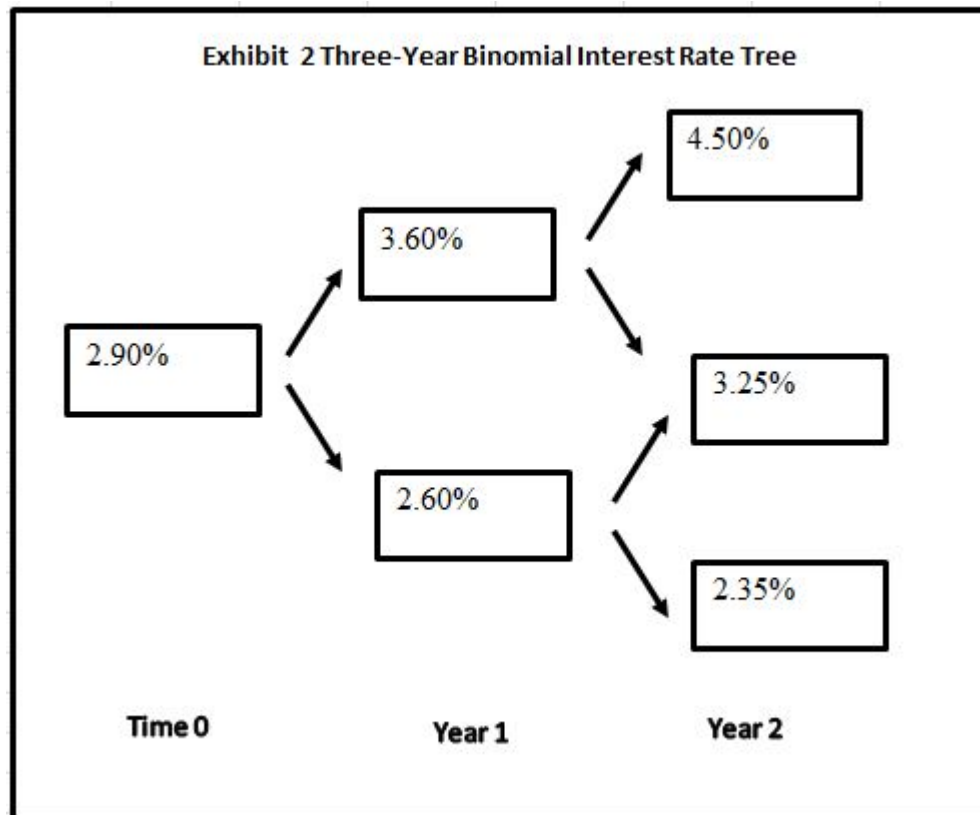
Comment 1: In the valuation process, the interest rate tree generates cash flows that are interest rate dependent; but does not provide the interest rates used to discount those cash flows.

Comment 2: Two assumptions must be made to create a binomial tree. The first is an interest rate model such as a lognormal model of interest rates. The second is a volatility of interest rates.

Comment 3: Volatility can be measured relative to the current level of rates. By using a lognormal distribution, interest rate movements are proportional to the level of rates and are bounded at the low end by zero.

Annisquam asks Hake to use a binomial interest rate tree to calculate the value of a bond. He tests the module using a three-year, \$100 par value, 4% annual pay coupon bond and the data in Exhibit 2.

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Annisquam tells Hake that he needs to calibrate the binomial interest rate tree to match a term structure of interest rates. Hake wants to better understand this process and asks Annisquam to describe it. Annisquam says, “Calibrating an interest rate tree requires an iterative process that ensures that the upper and lower rates are consistent with the volatility assumption, the interest rate model, and the observed market value of the benchmark bond. The cash flows of the bond are discounted using the interest rate tree, and if this doesn’t produce the correct price, another pair of forward rates is selected and the process is repeated.”

Annisquam then develops a model that compares the value of a bond determined using a binomial interest rate tree to its value determined using spot rates. The bond he selects for the comparison is non-benchmark, option-free, has five years to maturity and an annual-pay coupon rate of 3%. The coupon rate is below the coupon rate of the benchmark bond. The yield curve is currently downward sloping. The output of Annisquam’s model shows that the spot rates generate a value equal to the market price of the bond, but the interest rate tree methodology produces a higher value.

Annisquam wants Hake to develop a program for pricing securities that are interest rate path dependent, such as mortgage-backed securities (MBS). He believes that using the Monte Carlo method and employing 2,000 simulations will provide an average present value across all scenarios equal to the actual market value of the securities. Hake runs a simulation and uses it to value a benchmark bond. He finds that the value generated does not equal the market price of the bond.

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43. Based on the market price of the Tangoran government bond and the interest rates in Exhibit 1, what profitable arbitrage opportunity should Hake's algorithm *most likely* identify?
- A. Buying the Year 1 and Year 2 strips and selling the Year 3 strip
 - B. Buying the bond and selling the strips
 - C. Buying the strips and selling the bond
44. Which of Annisquam's comments regarding binomial interest rate trees is *least likely* correct?
- A. Comment 1
 - B. Comment 2
 - C. Comment 3
45. Using the backward induction method and the data in Exhibit 2, the value of the bond Hake has been asked to value is *closest* to:
- A. 101.069
 - B. 102.532
 - C. 101.584
46. Is Annisquam *most likely* correct in regard to his comments on calibrating a binomial interest rate tree?
- A. Yes
 - B. No, he incorrectly describes the iterative process
 - C. No, he is incorrect regarding the interest rate used
47. Assuming Annisquam's spot rate valuation is correct, why does his model *most likely* produce a different result?
- A. The model is incorrect because both methodologies should value the bonds equally.
 - B. He is valuing a non-benchmark bond.
 - C. The yield curve is downward sloping.
48. To correct the problem Hake encounters when using a Monte Carlo simulation, he would *most likely*:
- A. adjust the volatility assumption.
 - B. increase the number of simulations.
 - C. add a constant to all interest rates on all paths.

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Fixed Income – Scott

Halstead Capital Advisers (Halstead) is an investment advisory firm that specializes in taxable fixed-income investing. Its clients consist of medium-sized foundations and endowments that select outside managers, such as Halstead, after having formulated their investment policy and asset allocation targets.

Charles Scott, Halstead's chief investment strategist, and Catherine Bird, a quantitative analyst, meet to discuss a research report that Bird is producing. The report will address various fixed-income investing topics, including investment strategies, credit market spreads, and yield curve movements.

Bird is analyzing a newly issued US Treasury bond with a five-year maturity and a 7.00% coupon. For long-term investors that buy this US Treasury bond and hold it to maturity, Bird is assessing whether the realized return will match its current 7.00% yield to maturity. Her analysis is based on an expectation that the forward path of interest rates will follow the current spot rate curve. Current spot rates and extrapolated one-year forward rates are provided in Exhibit 1.

Exhibit 1 Spot and Forward Interest Rates		
Year	Spot Rate	Forward Rate
1	3.00%	
2	4	5.01%
3	5	7.03
4	6	9.06
5	7	11.1

For another investor who may sell prior to maturity, Scott states that the future value of this Treasury bond is a function of projected spot rates relative to the forward curve. Bird agrees and says, "Let's assume that an investor purchases this US Treasury bond at par, to yield 7.00% to maturity. He then holds the bond for two years, at which time the one-year, two-year, and three-year spot interest rates are each assumed to equal 8.00%." Bird determines that the expected return for the initial two-year holding period would equal 4.42%.

Scott recognizes that this US Treasury bond may not be suitable for investors who want zero reinvestment risk. He suggests that an alternative instrument is a US Treasury zero-coupon note. It is newly issued, with a five-year term, and priced at \$71.30 (\$100.00 face value) to yield 7.00% to maturity. Scott says that some investors may purchase this Treasury zero-coupon note today and hold it for five

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years to maturity. Scott continues by stating that other investors may purchase this Treasury zero-coupon note in two years and then hold it for three years to maturity. Scott asks Bird to determine the forward rate that would cause investors to be indifferent about either purchasing the Treasury zero-coupon note today or purchasing it two years from today.

Scott reminds Bird to include an update on credit instruments. He provides details on a newly issued zero-coupon bond by Coores Corporation, rated A1/A+, with five years to maturity priced to yield 7.30% to maturity. This credit typically trades in line with high-quality financial institutions and corporate issuers. Current market rates are 7% for the five-year risk-free spot rate, and the five-year swap spread is 0.30%.

Bird proposes to review other credit spread indicators that measure credit and liquidity risk for money market securities, general creditworthiness of individual debt issuers, and counterparty risk. Bird offers the following statements about measures of credit risk.

- Statement 1:** Z-spread represents the difference between the yield on credit bonds and the implied spot yield curve.
- Statement 2:** Libor-OIS (overnight index swaps) spread represents the difference between Libor and corporate bond spreads.
- Statement 3:** TED (Treasury-eurodollar) spread represents the difference between Libor and overnight bank lending rates.

Scott asks Bird to evaluate the impact of yield curve movements on fixed-income securities. Bird constructs a yield curve factor model that describes three independent yield curve movements. The yield curve movements are shown in Exhibit 2.

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Exhibit 2 Yield Curve Movements					
Time to Maturity	One Year	Two Years	Three Years	Four Years	Five Years
Factor 1	0.75%	1.10%	1.62%	2.27%	3.03%
Factor 2	-0.47	1.03	2.05	1.02	-0.45
Factor 3	0.98	0.99	1	1.01	1.02

49. Based on the data provided in Exhibit 1 and assuming that Bird's interest rate expectation materializes, the realized return for the US Treasury bond if held to maturity would *most likely* be:
- A. less than the yield to maturity.
 - B. equal to the yield to maturity.
 - C. greater than the yield to maturity.
50. Based on the data provided in Exhibit 1 and considering Bird's assumptions regarding an investor who purchases the US Treasury bond and holds it for two years, the US Treasury bond is currently *most likely*:
- A. undervalued.
 - B. overvalued.
 - C. fairly valued.
51. Using the information provided in Exhibit 1, the forward rate at which an investor would be indifferent to purchasing the US Treasury zero-coupon note today or two years from today is *closest to*:
- A. 11.10%.
 - B. 7.00%.
 - C. 9.05%.
52. Using the information provided, is the Coores Corporation bond *most likely* mispriced?
- A. Yes, because of the difference between the swap rate and the yield to maturity
 - B. No
 - C. Yes, because of the difference between the swap rate and the spot rate

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53. Which of Bird's statements regarding measures of credit risk is *most likely* correct?
- A. Statement 1
 - B. Statement 2
 - C. Statement 3
54. Using the information provided in Exhibit 2, the movements characterized by Factor 1, Factor 2, and Factor 3, respectively, are *most likely*:
- A. level, steepness, and curvature.
 - B. curvature, level, and steepness.
 - C. steepness, curvature, and level.

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Alternative Investments -Schulman

Zoe Schulman is an alternative investments research analyst with Principal Investments, LLC, a wealth management firm. In the past, the account managers at Principal were free to select real estate investments without referring to a formal buy list, leading to inconsistent returns across client portfolios. To ensure a more consistent approach, the firm would like to create an approved list, which would provide a source of investment selections for all client portfolios.

From the investments already held in client portfolios, Schulman identifies the three largest REIT holdings and asks the account managers to justify why these REITs were selected considering the current economic cycle. She compiles Exhibit 1 and presents it in a report to her manager, Holden Dwelley.

Exhibit 1 Characteristics of Major REIT Holdings and Account Managers' Justifications			
REIT	Sector	Region	Justification
1	Industrial	Midwest United States	The industrial sector has a large labor force and is most sensitive to new job creation.
2	Office	Southwest United States	Although recent news of rising GDP is beneficial to all sectors, the better-than-expected jobs report further enhances prospects for the office property sector.
3	Hospitality	Northeast United States	This sector has low sensitivity to new job creation and is resilient during times of economic contraction.

Schulman determines that all three REITs use the historical cost method in their accounting statements. In her analysis, she calculates and reports net asset value per share (NAVPS), instead of book value per share (BVPS), as an absolute valuation metric and provides the following rationale for this approach being her preferred one:

Reason 1: NAVPS accounts for the value of property not currently generating revenue.

Reason 2: NAVPS includes the added value of the management of the REIT.

Reason 3: NAVPS reflects the market value of the property portfolio rather than often stale historical cost values.

Schulman collects financial data for all three REITs to calculate NAVPS. Exhibit 2 presents select data that she uses for her evaluation of REIT 1.

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Exhibit 2 Selected Data for REIT 1	
<i>Pro forma</i> cash net operating income (NOI) for last 12 months	\$320 million
Cash and equivalents	\$50 million
Land held for future development	\$70 million
Accounts receivable	\$25 million
Prepaid/other assets (excluding intangibles)	\$7 million
Total debt	\$1,700 million
Other liabilities	\$200 million
Shares outstanding	75,000
Estimated growth in NOI	5.00%
Capitalization rate	7.00%

Schulman submits a first draft of her report to Dwelley. He notes that she has failed to consider real estate operating companies (REOCs) in current client portfolios for inclusion on the approved list. She justifies the omission with the following reasoning:

- REOCs are more constrained in their use of leverage than REITs.
- REOCs are far less commonly traded in the United States than REITs.
- REOCs have more restrictive real estate investment choices.

When editing the report, Dwelley questions Schulman's reliance on NAVPS over a dividend discount model (DDM) and notes the following characteristics of these valuation measures:

Characteristic 1: DDM reflects all earned income, whereas NAVPS only reflects income that is retained by the property management company.

Characteristic 2: NAVPS is based solely on historical revenue and does not reflect upcoming income growth expectations.

Characteristic 3: DDM uses discount rates consistent with the required rate of return for public equity investment.

For a more comprehensive analysis, Schulman's report also presents relative valuation measures, such as the ratio of price to funds from operations (P/FFO). Selected information for REIT 3 is provided in Exhibit 3

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Exhibit 3 Selected Information for REIT 3	
Property net operating income	\$700 million
General and administrative expenses	<u>\$70 million</u>
EBITDA	\$630 million
Depreciation and amortization	\$30 million
Net interest expense	<u>\$170 million</u>
Net income available to common	\$430 million
Non-cash (straight-line) rent adjustment	\$15 million
Recurring maintenance-type capital expenditures and leasing commissions	\$33 million
Number of shares outstanding	140
Price per share	\$49

55. Which of the account managers' justifications in Exhibit 1 regarding the selection of each of the three REITs is *most likely* correct?
- A. The justification for REIT 3
 - B. The justification for REIT 2
 - C. The justification for REIT 1
56. Which of Schulman's reasons regarding her preferred approach to valuing REITs is *most likely* correct?
- A. Reason 3
 - B. Reason 2
 - C. Reason 1
57. Using the data in Exhibit 2, the NAVPS of REIT 1 is *closest* to:
- A. \$39.76.
 - B. \$37.65.
 - C. \$40.69.

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58. Which of Schulman's justifications for omitting REOCs from the approved list is *most likely* correct?

- A. Her justification regarding investment restrictions
- B. Her justification regarding trading activity
- C. Her justification regarding leverage

59. Which of Dwelley's characteristics of the DDM and NAVPS methods is *most likely* correct?

- A. Characteristic 1
- B. Characteristic 3
- C. Characteristic 2

60. Using Exhibit 3, the P/FFO for REIT 3 is *closest* to:

- A. 16.7.
- B. 13.5.
- C. 14.9.

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